

Aditya Kaushikbhai Rathod

B.Tech. in Electrical Engineering | Indian Institute of Technology, Jodhpur

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EDUCATION

Degree/Certificate	Institute/School	CGPA/Percentage	Year
B.Tech. EE	Indian Institute of Technology, Jodhpur	7.38/10	2024-Present
Senior Secondary	Tapovan School, Rajkot	88.8%	2024

EXPERIENCE

- Contributor - plink (Peer-to-Peer File Transfer)** May 2025 - Nov. 2025
Python, Systems Development, Peer-to-Peer Networking, AES-256 Encryption [GitHub](#)
 - Built a secure peer-to-peer file transfer tool enabling direct device-to-device sharing without centralized servers.
 - Engineered modular networking for direct connection, UPnP, NAT hole punching, and role reversal.
 - Fortified transfers with AES-256 end-to-end encryption, secure key exchange, and file verification.
 - Optimized large-file reliability with smart chunking and cross-platform CLI workflows for Windows, macOS, and Linux.

PROJECTS

- Adaptive Traffic Signal Control using Reinforcement Learning** May 2026
Python, PyTorch, PPO, Stable-Baselines3, Gymnasium, TensorBoard, NumPy [GitHub](#)
 - Developed an adaptive traffic signal optimization system using PPO, actor-critic methods, and a custom Gymnasium simulation environment.
 - Designed vehicle-density and waiting-time state features, stochastic traffic arrivals, and reward engineering strategies to reduce congestion and improve throughput.
 - Trained a PPO agent for 500,000+ simulation timesteps using 16 vectorized environments with TensorBoard diagnostics for reward and policy monitoring.
 - Evaluated learned policies against random-control baselines, achieving 2341.5 final test reward and 2371.47 mean evaluation reward with 25.57 standard deviation.
- Quadcopter Drone with Onboard Vision** Aug. 2025 - Mar. 2026
NVIDIA Jetson Orin NX, YOLO, Computer Vision, Edge AI, Real-Time Inference [GitHub](#)
 - Built and flight-tested a manually piloted quadcopter platform, sustaining approximately 20 minutes of flight time on a 3S battery.
 - Integrated NVIDIA Jetson Orin NX onboard compute and deployed a YOLO object-detection pipeline on live camera feeds for edge AI perception.
 - Achieved 10-14 FPS real-time inference while balancing detection accuracy, onboard compute limits, and power constraints.
 - Led the onboard vision and compute subsystem while supporting flight-control integration and autonomy workflows.
- Edge AI Face Recognition System** Feb. 2025
ESP32-CAM, FTDI232, Edge Impulse, TinyML, On-Device Inference, Computer Vision [Project](#)
 - Developed a low-cost Edge AI face-recognition system using ESP32-CAM, Edge Impulse, and LCD output for portable real-time vision.
 - Trained and deployed an on-device machine-learning model for facial recognition without cloud dependence.
 - Executed dataset collection, labeling, and Arduino IDE deployment, achieving approximately 90% internal-test accuracy.
 - Optimized the prototype for low-power, real-time inference on resource-constrained embedded hardware.
- Autonomous Maze-Solving Robot** Jan. 2025
Arduino Uno, PID Control, Ultrasonic Sensors, Motor Driver, Differential Drive, Embedded C/C++ [GitHub](#)
 - Engineered a PID-based autonomous wall-following robot using Arduino and HC-SR04 ultrasonic sensors for real-time distance tracking.
 - Implemented wall-following, obstacle-avoidance, and turn-decision logic for reliable navigation through constrained paths.
 - Optimized anti-windup PID feedback for dynamic speed regulation and differential-drive motor control through the L298N driver.
 - Instrumented serial-monitor debugging to analyze sensor readings, PID error, motor speeds, and wall-side selection during testing.

TECHNICAL SKILLS

Languages:	C, C++, Python
AI/ML Frameworks:	PyTorch, Stable-Baselines3, Gymnasium, TensorBoard, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV
ML & Vision:	Reinforcement Learning, PPO, Policy Gradients, Actor-Critic Methods, Deep Learning, Neural Networks, CNNs, Computer Vision, YOLO, TinyML, Edge AI
Systems:	Jetson Orin NX, ESP32-CAM, Arduino, Raspberry Pi, PID Control, MAVLink, Peer-to-Peer Networking, AES-256 Encryption
Tools:	Git, GitHub, Linux, Jupyter, MATLAB, Arduino IDE, Edge Impulse, Figma
Core CS:	Data Structures and Algorithms, Object-Oriented Programming, Multithreading, Debugging

RELEVANT COURSEWORK

- Pattern Recognition and Machine Learning, Data Structures and Algorithms, Introduction to Computer Science, Probability, Statistics and Stochastic Processes, Signals & Systems, Digital Design

CAMPUS INVOLVEMENT & LEADERSHIP

- Secretary**, Designers - The Design Club, Indian Institute of Technology, Jodhpur Aug. 2025 - Apr. 2026
- Student Guide**, SWC - Student Wellbeing Committee, Indian Institute of Technology, Jodhpur Aug. 2025 - Apr. 2026
- Core member**, Inside - The Game-dev Society, Indian Institute of Technology, Jodhpur Aug. 2025 - Apr. 2026
- Assistant Head**, Ignus'26, Indian Institute of Technology, Jodhpur Jan. 2026 - Feb. 2026
- Assistant Head**, Prometeo'26, Indian Institute of Technology, Jodhpur Oct. 2025 - Jan. 2026
- Assistant Head**, Varchas'25, Indian Institute of Technology, Jodhpur Sep. 2025 - Nov. 2025